ISE 333: User Interface Development (Design)
Chapter 15: Information Search
Chapter 16: Data Visualization

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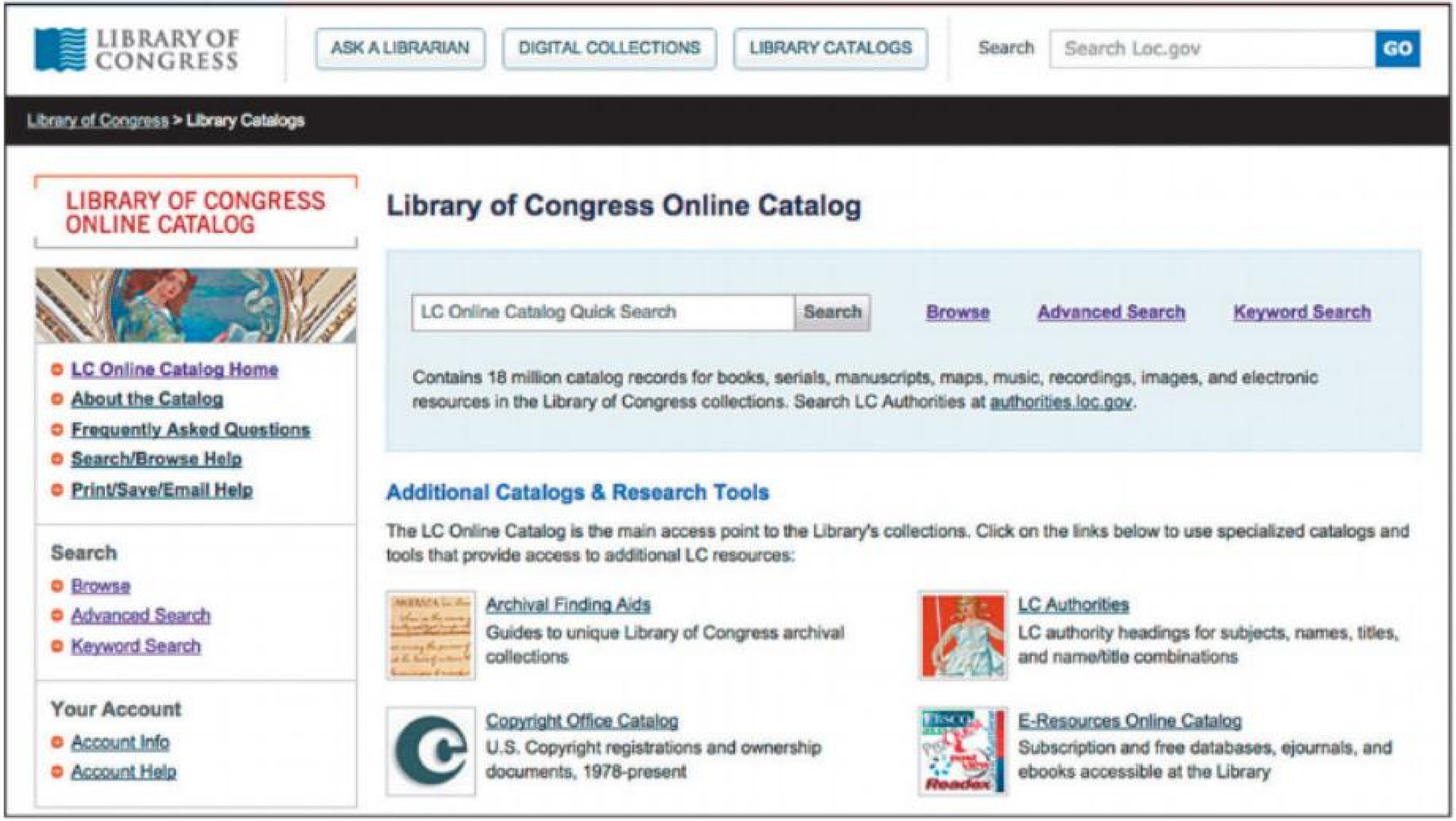
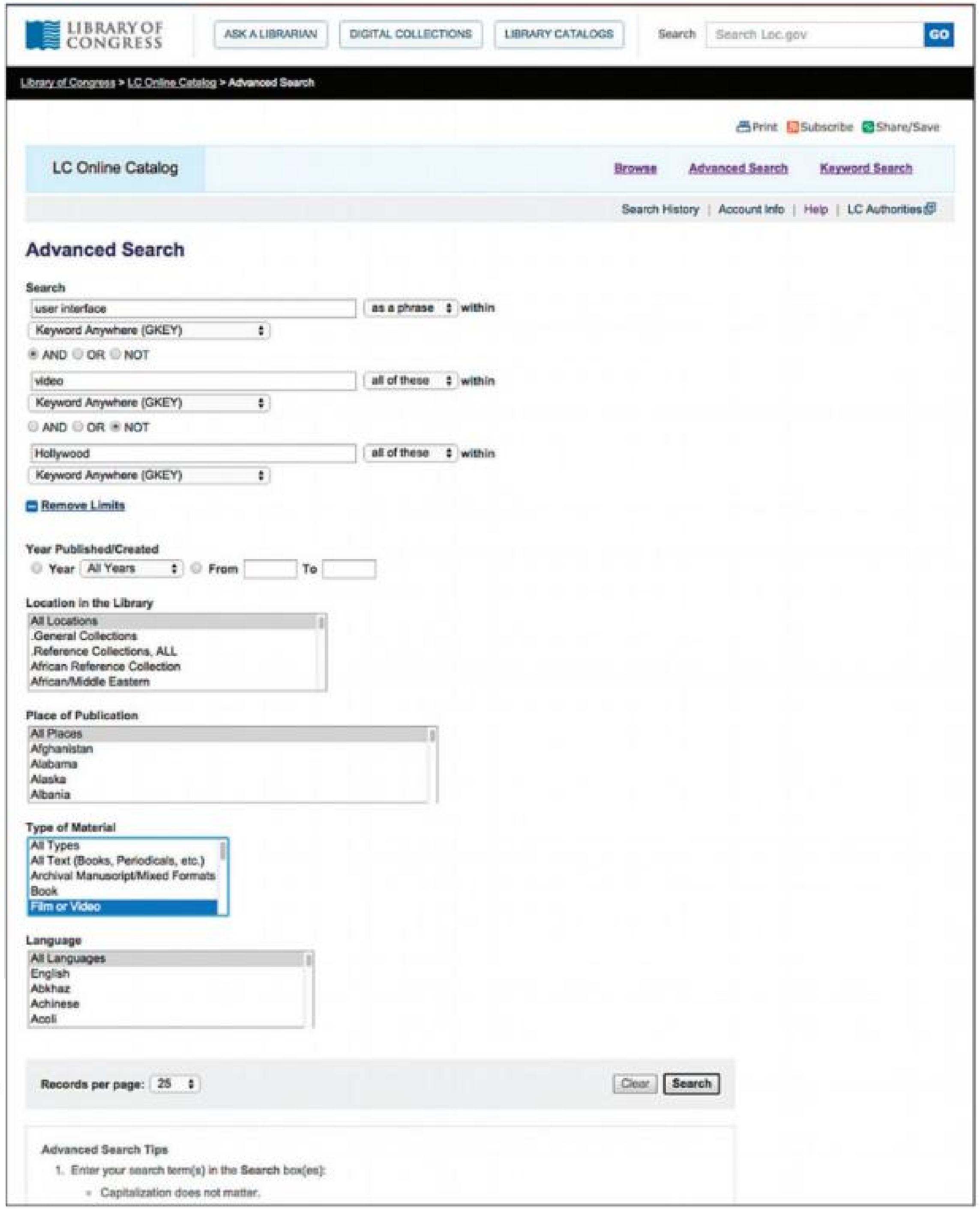
Information Search

The home page of the U.S. Library of Congress Online Catalog (https://catalog.loc.gov) shows the simple search box prominently placed at the top of the page and provides alternative means of finding items of interest in the diverse collections. Advanced search interfaces are provided to accommodate experienced searchers.



Information Search



The advanced search interface of the U.S. Library of Congress Online Catalog (https://catalog.loc.gov). The entire page is now dedicated to search controls and tips. Using check boxes, text fields, and menus, users can compose Boolean queries, restrict the search scope to a subset of the collections, and apply filters based on metadata. Regular users sign up for an account to save results and keep a search history to facilitate re-finding.

Information Search

- Five-Stage Search Framework
  - Formulation
    - Expressing the search
  - Initiation of action
    - Launching the search
  - Review of results
    - Reading messages and outcomes
  - Refinement
    - Formulating the next step
  - Use
    - Compiling or disseminating insight

1. Formulation

- Use simple and advanced search.
- Limit the search using structured fields such as year, media, or location.
- Recognize *phrases* to allow entry of names, such as "George Washington."
- Permit *variants* to allow relaxation of search constraints (e.g., phonetic variations).
- Control the size of the initial result set.
- Use scoping of source carefully.
- Provide suggestions, hints, and common sources.

2. Initiation of action

- Explicit actions are initiated by buttons with consistent labels (such as "Search").
- Implicit actions are initiated by changes to a parameter and update results immediately.
- Guide users to successful or past queries with auto-complete.

3. Review of results

- Keep search terms and constraints visible.
- Provide an overview of the results (e.g., total number).
- Categorize results using metadata (by attribute value, topics, etc.).
- Provide descriptive previews of each result item.
- Highlight search terms in results.
- Allow examination of selected items.
- Provide visualizations when appropriate (e.g., maps or timelines).
- Allow adjustment of the size of the result set and which fields are displayed.
- Allow change of sequencing (alphabetical, chronological, relevance ranked, etc.).

4. Refinement

- Guide users in progressive refinement with meaningful messages.
- Make changing of search parameters convenient.
- Provide related searches.
- Provide suggestions for error correction (without forcing correction).

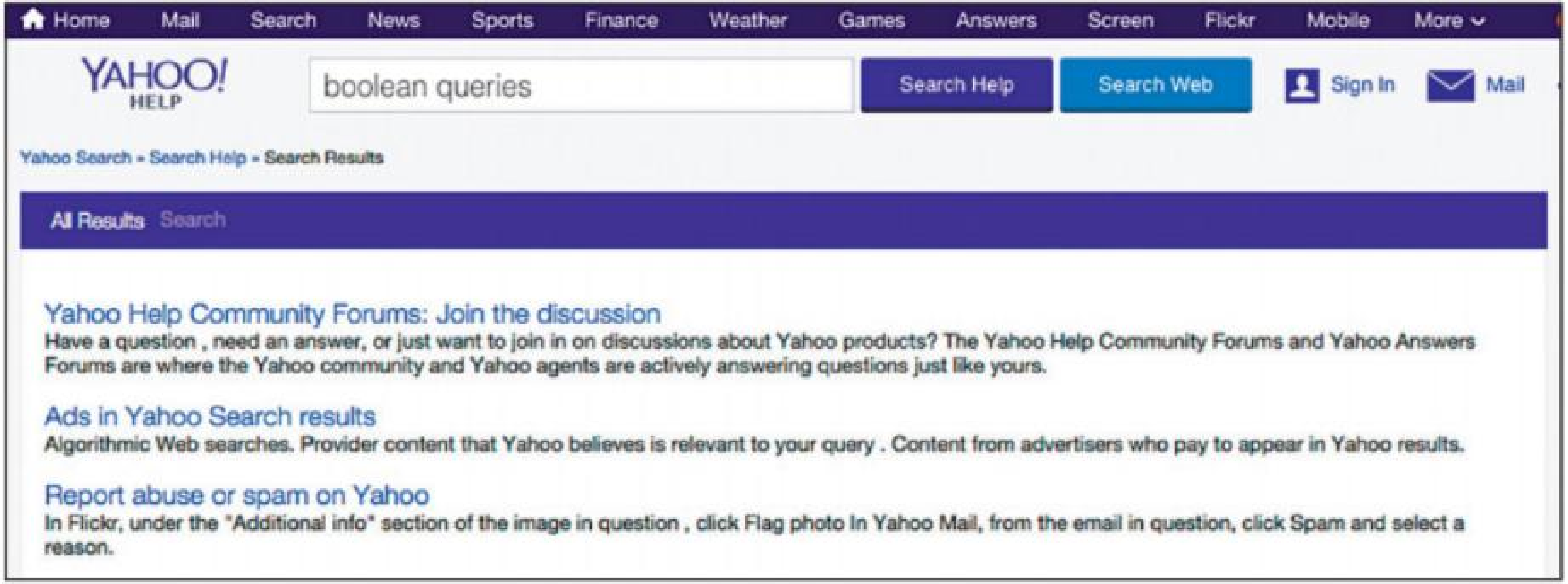
5. Use

- Imbed actions in results when possible.
- Allow queries, settings, and results to be saved, annotated, and sent to other applications.
- Explore collecting explicit feedback (ratings, reviews, like, etc.).

Information Search

- Formulation

The Yahoo! help search box has two buttons of different colors to search two different sources of information: purple for searching the help information and blue for searching the web. Pressing the purple button "scopes" the results to the help information only and shows results below a purple banner. Searching the web jumps to a different page (the normal search) that reuses the blue button color, helping users keep track of which source of information they are searching.



Information Search

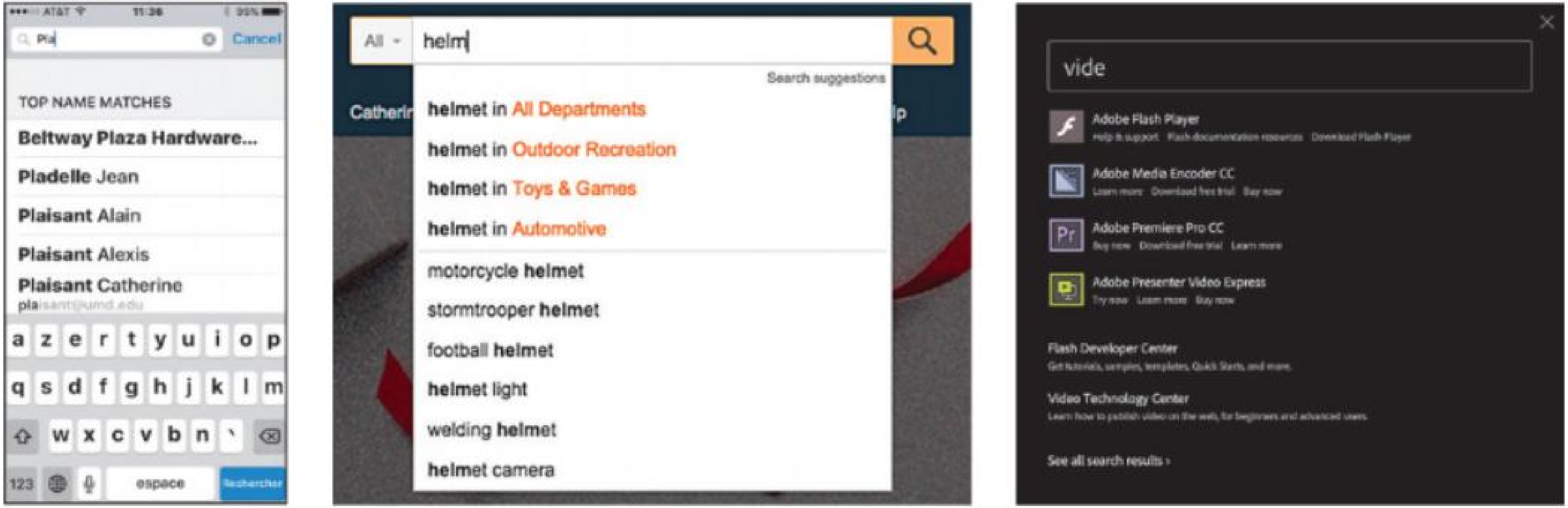
- Formulation

Auto-complete suggestions can speed data entry and guide users toward successful queries.

(a) In a mobile phone address book, typing one character filters the list to all names that contain that character, and the list is updated continuously as users type.

(b) Typing "helm" in Amazon's search box shows suggestions for "helmet light" or "welding helmet" but also suggestions to narrow the scope of the search to relevant departments.

(c) On the Adobe website, suggestions include products (e.g., typing the beginning of the word "video" suggests several video-editing tools).

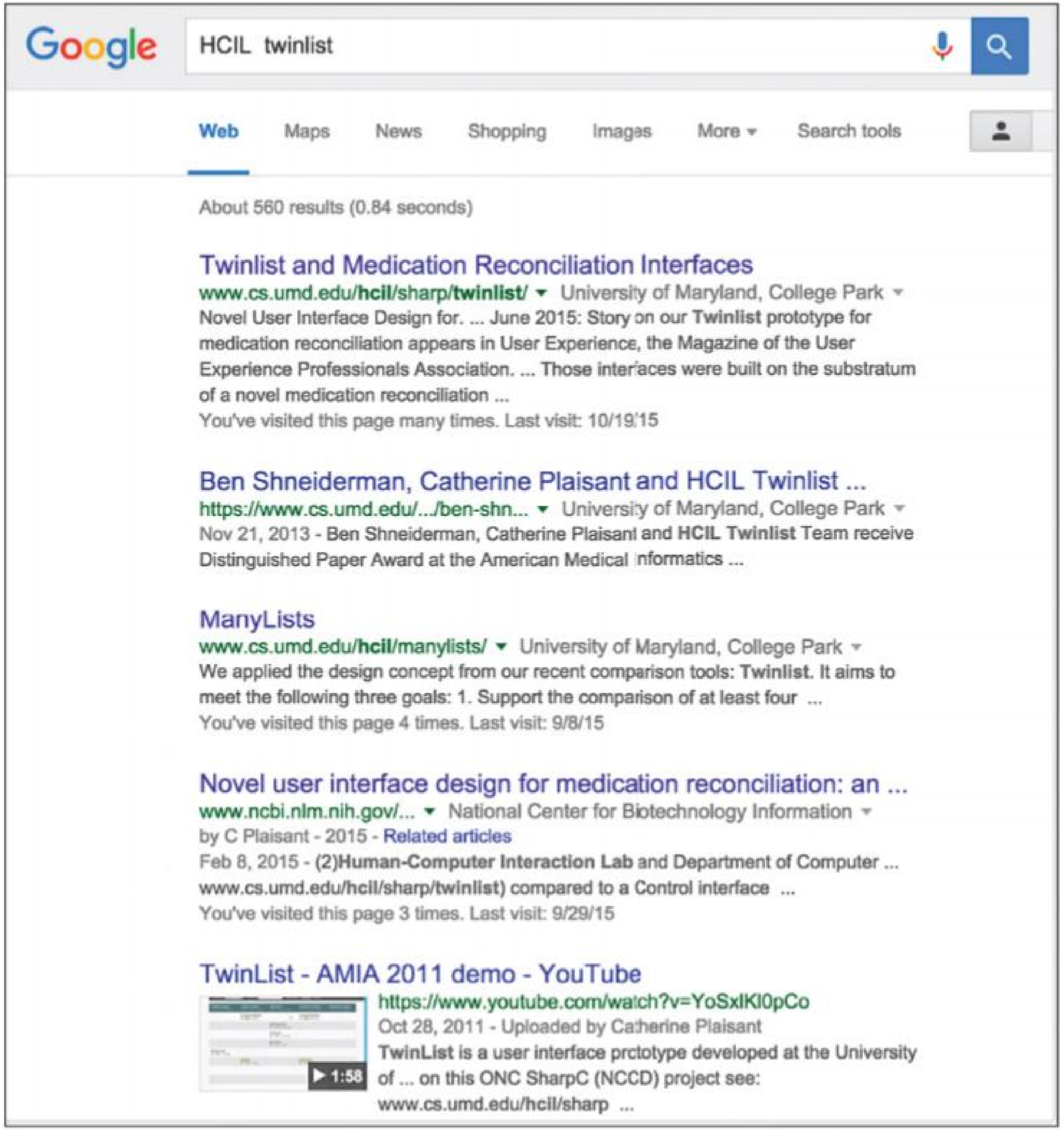


Information Search

- Initiation of action
  - Explicit initiation
    - Click on a search button (or press the Enter key on a keyboard) to initiate the search
  - Implicit initiation
    - Each change to any component of the formulation stage immediately produces a new set of search results
    - Dynamic queries enable users’ adjustment of query widgets to produce continuous updates

Information Search

- Review of results

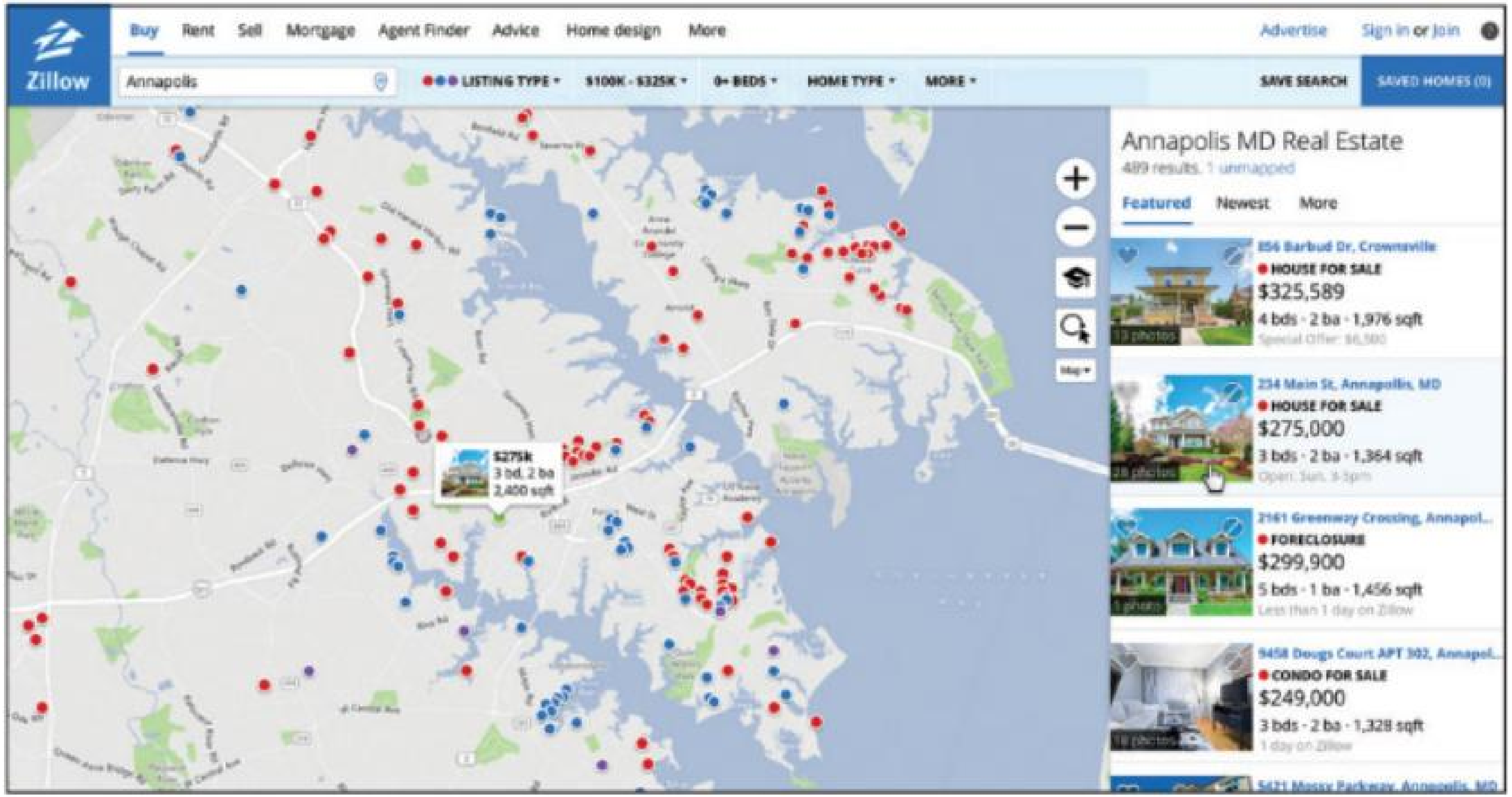


A Google Search result list. A summary is provided at the top (the total number of results). Each result includes preview information (or a snippet). Search terms are highlighted, including "Human-Computer Interaction Lab," which is the expanded variant of the search term "HCIL." The name of the top-level organization was added (here "National Center for Biotechnology Information") to help users judge the trustiness of the information.

Information Search

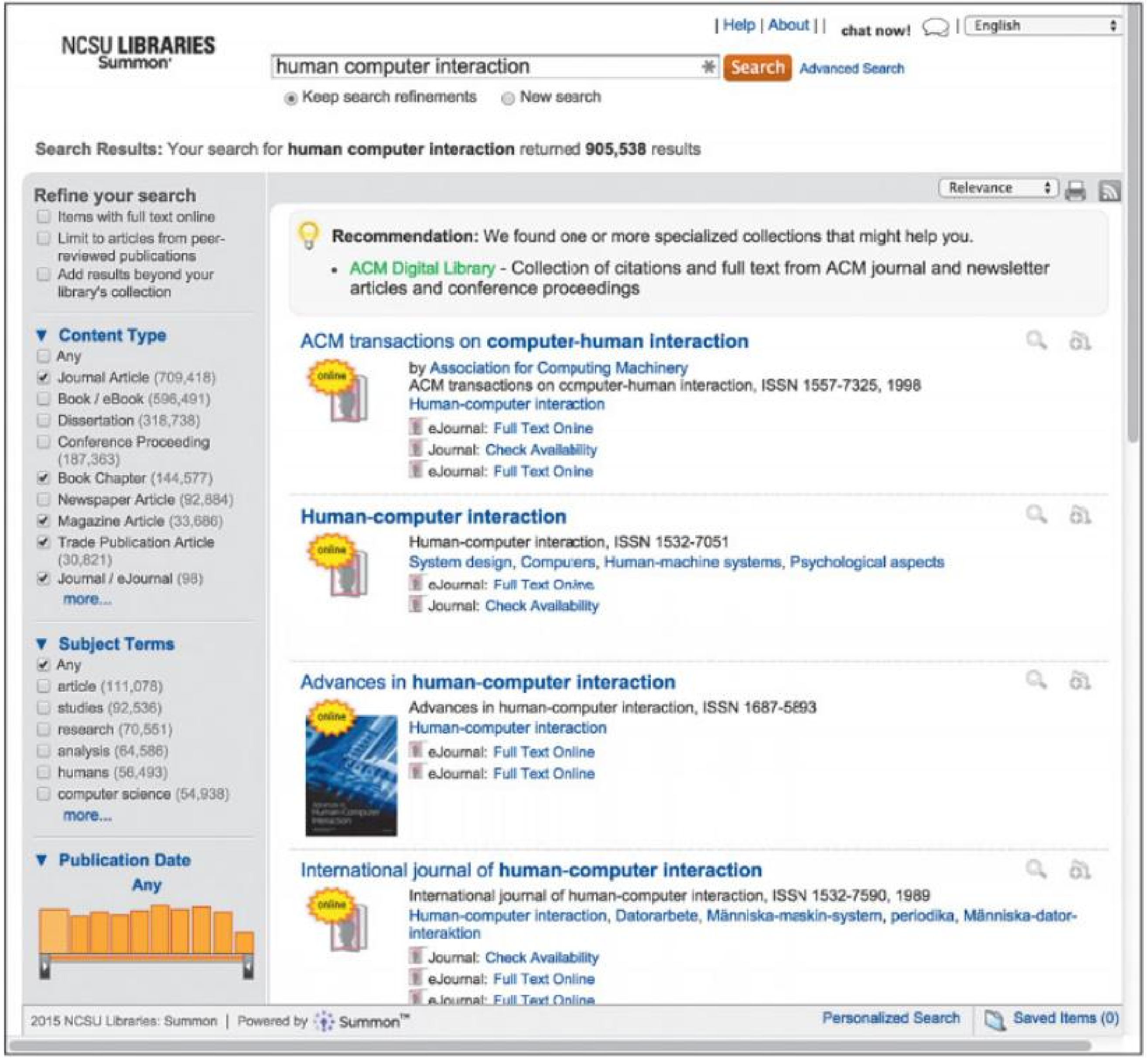
- Review of results

Searching for Annapolis on the real estate website Zillow returns a list of houses and dots displayed on a map. The two windows are coordinated; when the cursor hovers over a house in the result list, the location of the house is indicated on the map. A click on the house would bring all the details displayed in an overlapping window.



Information Search

- Review of results



A search for "human computer interaction" powered by Summon™ for a university library catalog returns a very large number of results. On the left, users can see the number of results for categories organized by content type, subject terms, or publication date. The box provides an overview of the results, reveals how the search was done (e.g., here the default search does not returns dissertations), and facilitates further refinement of the search. The menu at the upper right allows users to sort results by relevance or by date. Help is available with a "Chat now" button, which allows users to chat with a librarian (http://www.lib.ncsu.edu).

Information Search

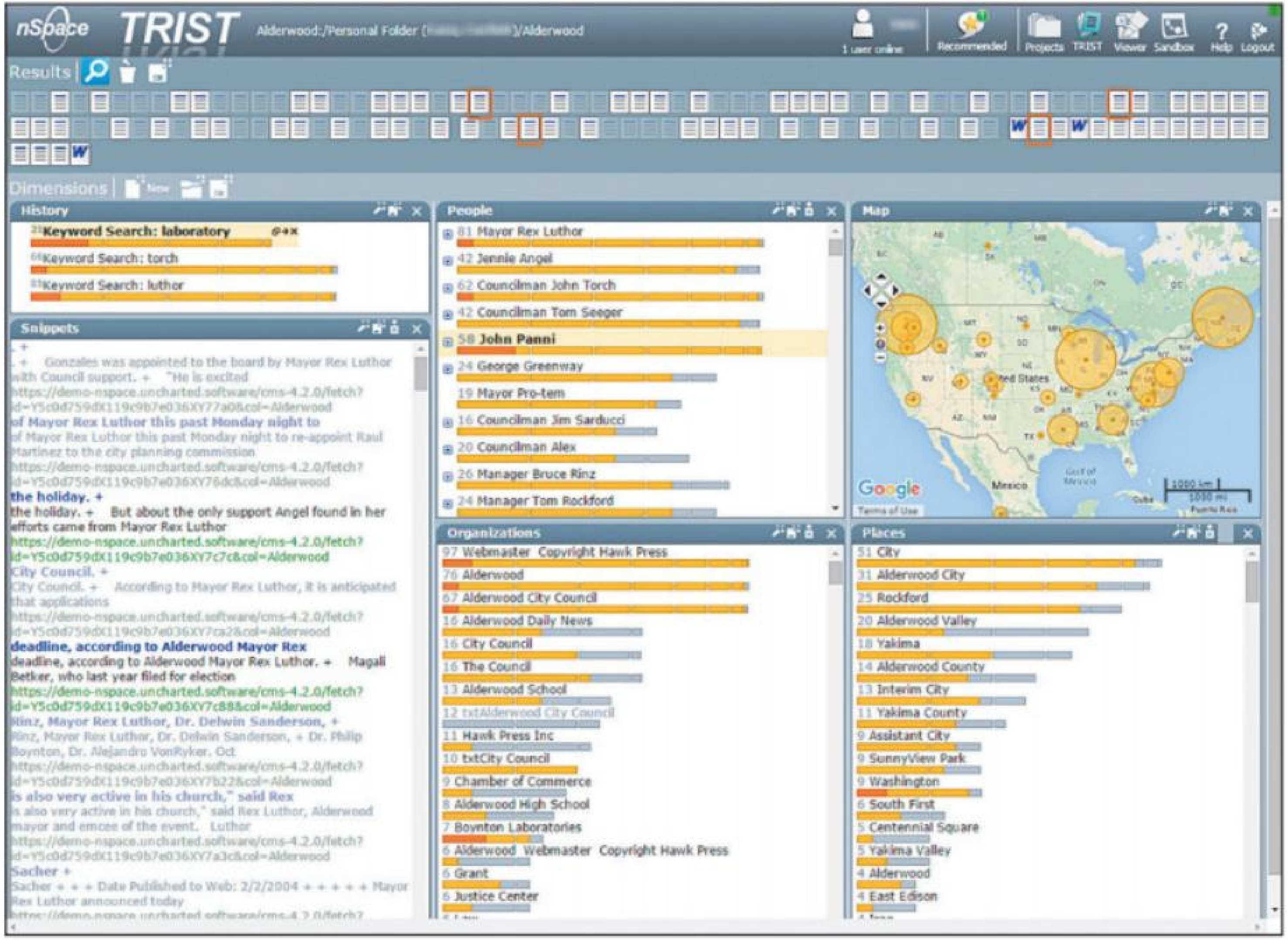
- Refinement
  - Search UIs can provide meaningful messages to explain search outcomes and to support progressive refinement
- Use

When possible (and important), provide information or simple actions without requiring users to leave the search results page. On the left, Google Search users get the answer to their safety-critical question at the top of the result list. On the right, Peapod shoppers looking for groceries can specify quantity and buy directly from the list of results after a search on "grapes."

Information Search

• Use

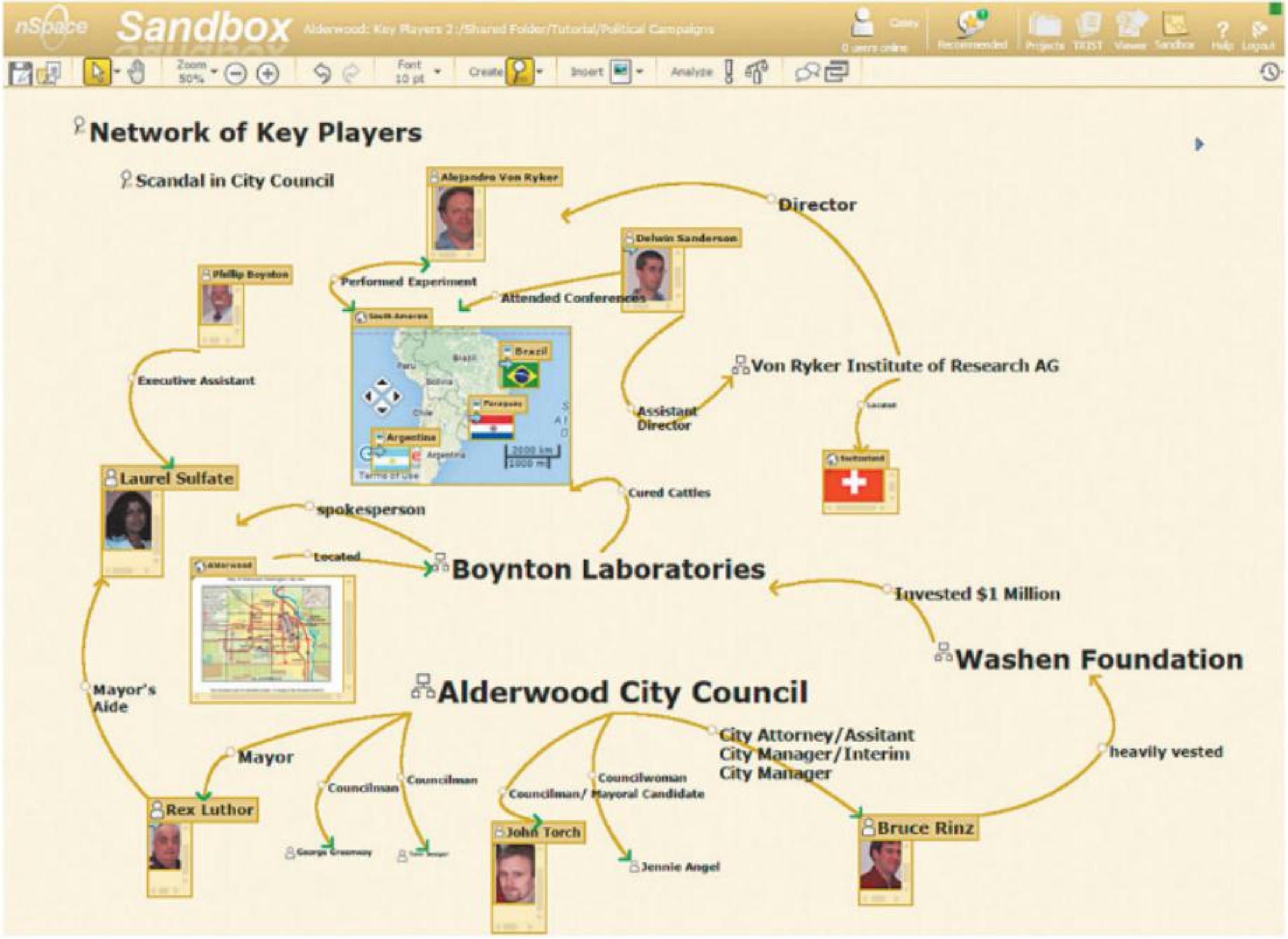
nSpace TRIST from Uncharted Software is used by analysts to produce evidence-based reports. For this (fictitious) criminal investigation, a user is reviewing a collection of documents (shown as icons at the top). The search history on the left shows three searches. Names, places, and organizations have been automatically extracted. Here the search term "Laboratory" and the person "John Panni" are selected. Snippets are shown on the bottom left. Analysts use TRIST to define dimensions of interest and to quickly identify documents of interest for use in a Sandbox (Fig. 15.10) (http://uncharted.software/nspace; Chien et al., 2008).



Information Search

- Use

nSpace Sandbox® from Uncharted Software™ allows multiple analysts to organize and present the evidence gathered from research. A variety of tools such as node and link diagramming, automatic source attribution, recursive evidence marshaling, and timeline construction provide support for analysis and reporting.



Information Search

- Miscellaneous Modalities
  - Command language based search
  - Natural language based search

- Miscellaneous Data
  - Image search
  - Video search
  - Audio search
  - Geographic information search

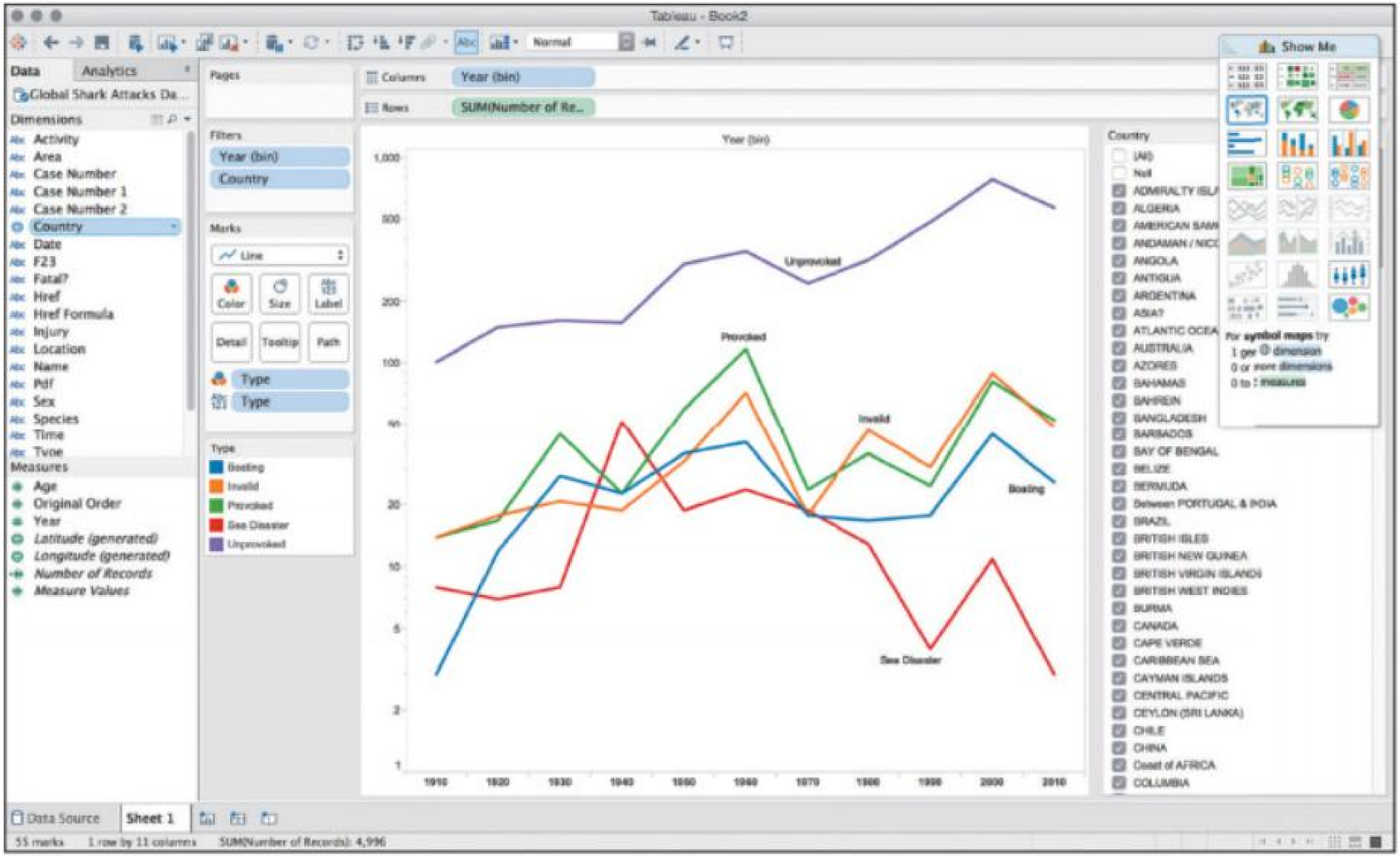
Data Visualization

- Tasks in Data Visualization

Task Categories	*Task Types*
Data and view specification	**Visualize** data by choosing visual encodings **Filter** out data to focus on relevant items **Sort** items to expose patterns **Derive** values of models from source data
View manipulation	**Select** items to highlight, filter, or manipulate **Navigate** to examine high-level patterns and low-level detail **Coordinate** views for linked exploration **Organize** multiple windows and workspaces
Process and provenance	**Record** analysis histories for revisitation, review, and sharing **Annotate** patterns to document findings **Share** views and annotations to enable collaboration **Guide** users through analysis tasks or stories

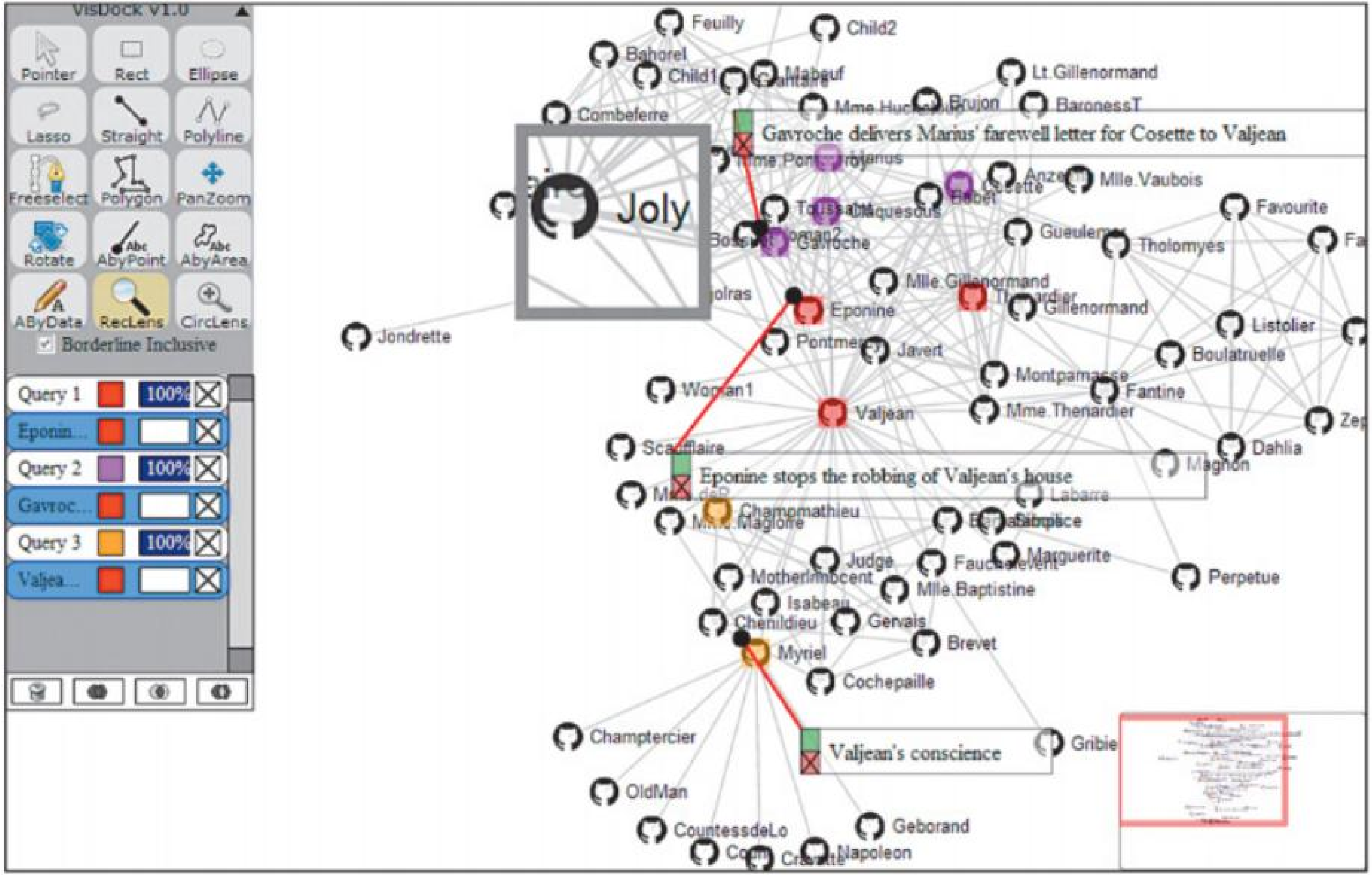
Data Visualization

Visualization palette (upper right) in the Tableau Desktop application for a dataset of shark attacks. The "show me" feature in the tool (Mackinlay et al., 2007) will automatically highlight the suitable charts that can be used for the selected data.



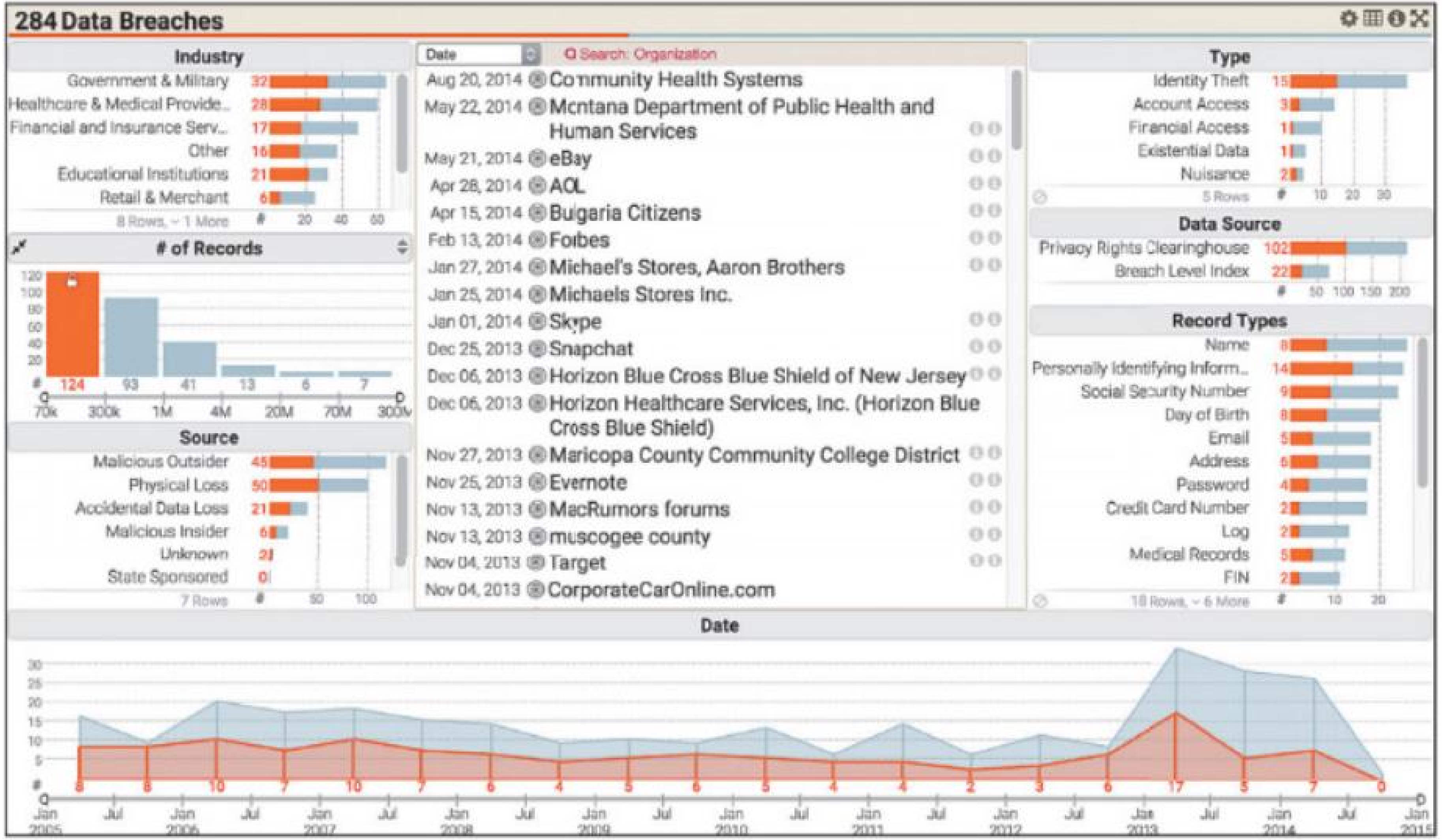
Data Visualization

Selection tools and data-aware annotations in an interactive node-link diagram representation of the social network for all of the characters in Victor Hugo's *Les Misérables*. Characters are linked together if they appear in the same chapter in the book. The textual annotations are connected to nodes using red lines and stay connected as the graph layout changes. The toolbar on the upper left is part of the VisDock toolkit and provides tools for annotation, navigation, and selection



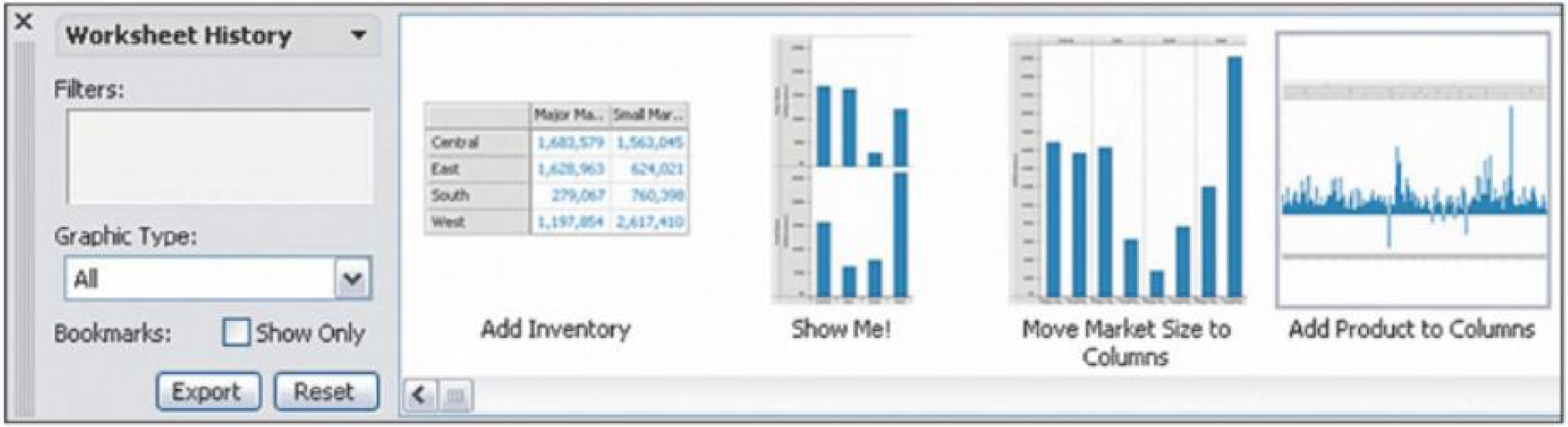
Data Visualization

Exploring 284 data breaches in the United States using Keshif (http://keshif.me/), a multi-view visualization tool that shows different aspects of the data in separate views (Yalcin, 2016). Selecting items in one view highlights them in others; for example, the user is currently hovering over the bar for “70k–300k” in the view titled “# of Records,” which causes those 124 breaches to also be highlighted in orange in other views, including in the timeline at the bottom.



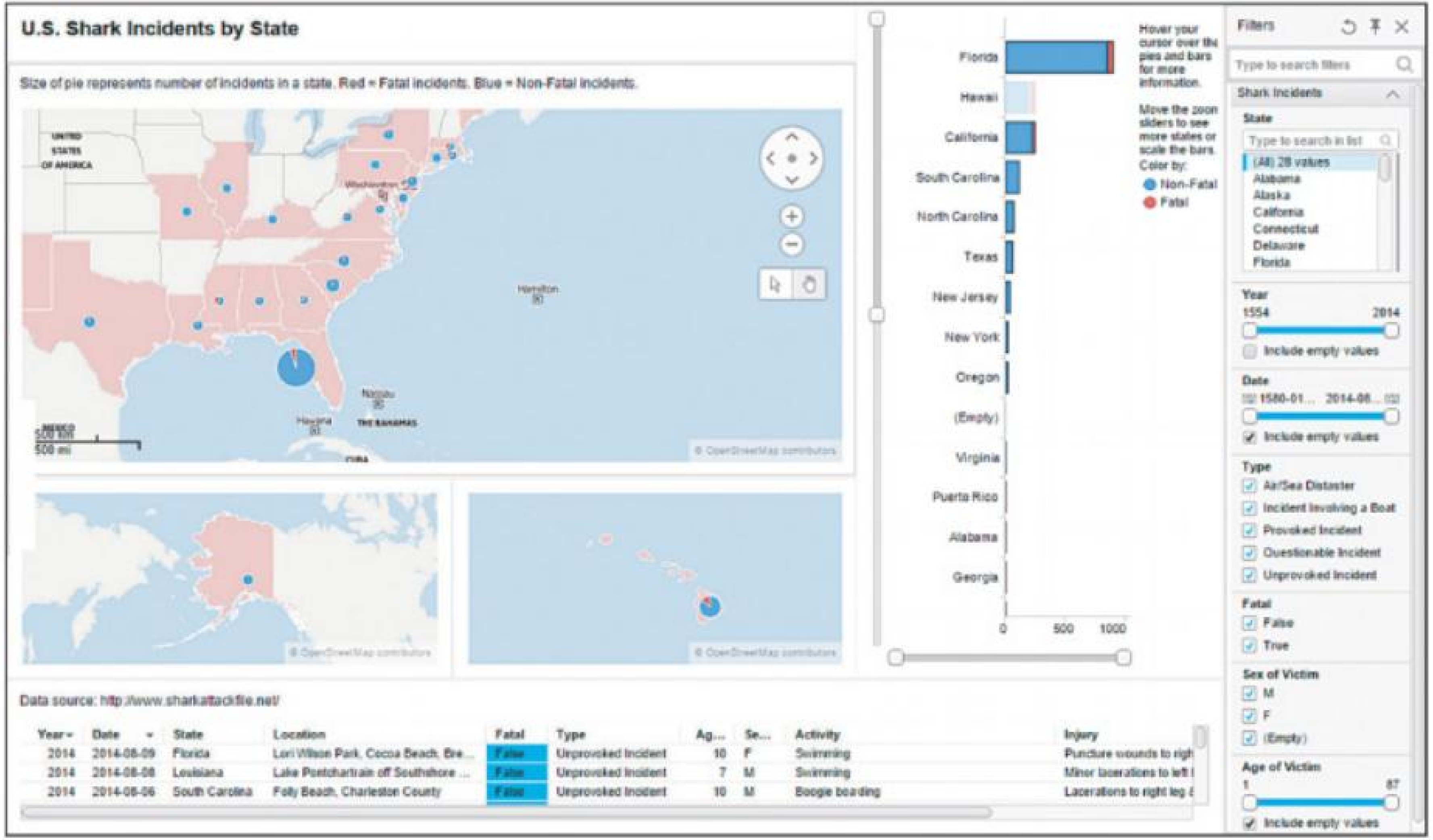
Data Visualization

Graphical history interface using thumbnails of previous visualization states organized in a comic-strip layout (Heer et al., 2008). The labels describe the actions performed.



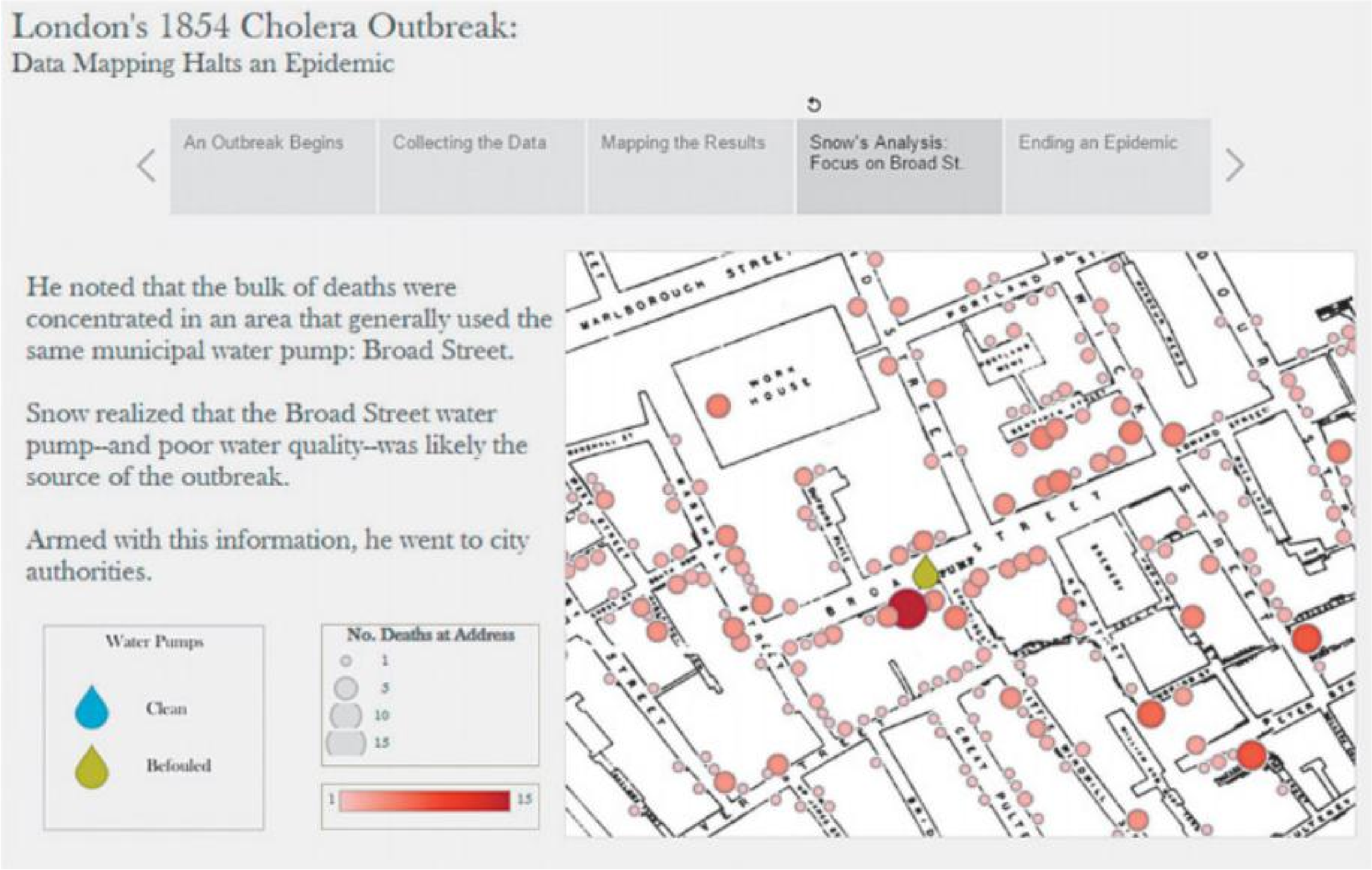
Data Visualization

Spotfire visualization dashboard of shark attacks published on the web. Users can interact with the dashboard, causing views to update dynamically. The tool also allows for application bookmarking (storing the state for specific insights) as well as sharing the analysis on social media platforms such as Facebook, Twitter, and LinkedIn.



Data Visualization

Web-based visualization of London's 1854 cholera outbreak showing physician John Snow's use of visualization to find its source. This visualization was created in Tableau using its Story Points feature, which allows users to build a narrative from data. The horizontal list of five boxes at the top of the display are the main points in the story, and viewers can be automatically guided through the story by moving to each point from left to right.



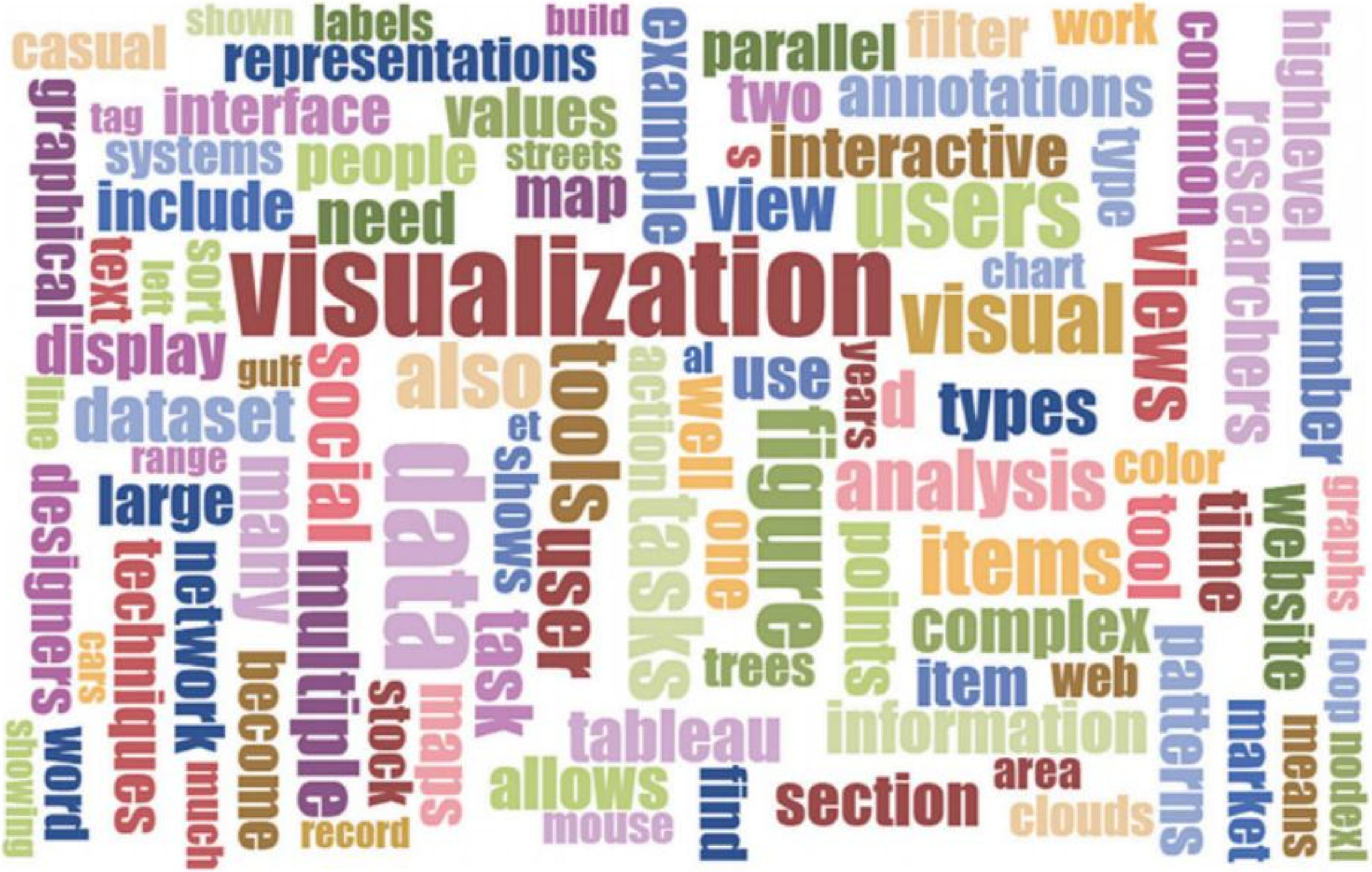
Data Visualization

- Visualization by Data Type

Data Type	*Visualization Techniques and Systems*
1-D linear	Tag clouds, Wordle, PhraseNets, parallel tag clouds
2-D space	Geographic information systems (GIS), self-organizing maps
3-D volume	Volume rendering, medical visualization, molecule visualization
Multi-dimensional	Tableau, parallel coordinates, scatterplot matrices
Temporal	Google Finance, EventFlow, LifeLines, TimeSearcher
Tree	Treemaps, degree of interest trees, space trees
Network	Node-link diagrams, adjacency matrices, NodeXL, Cytoscape

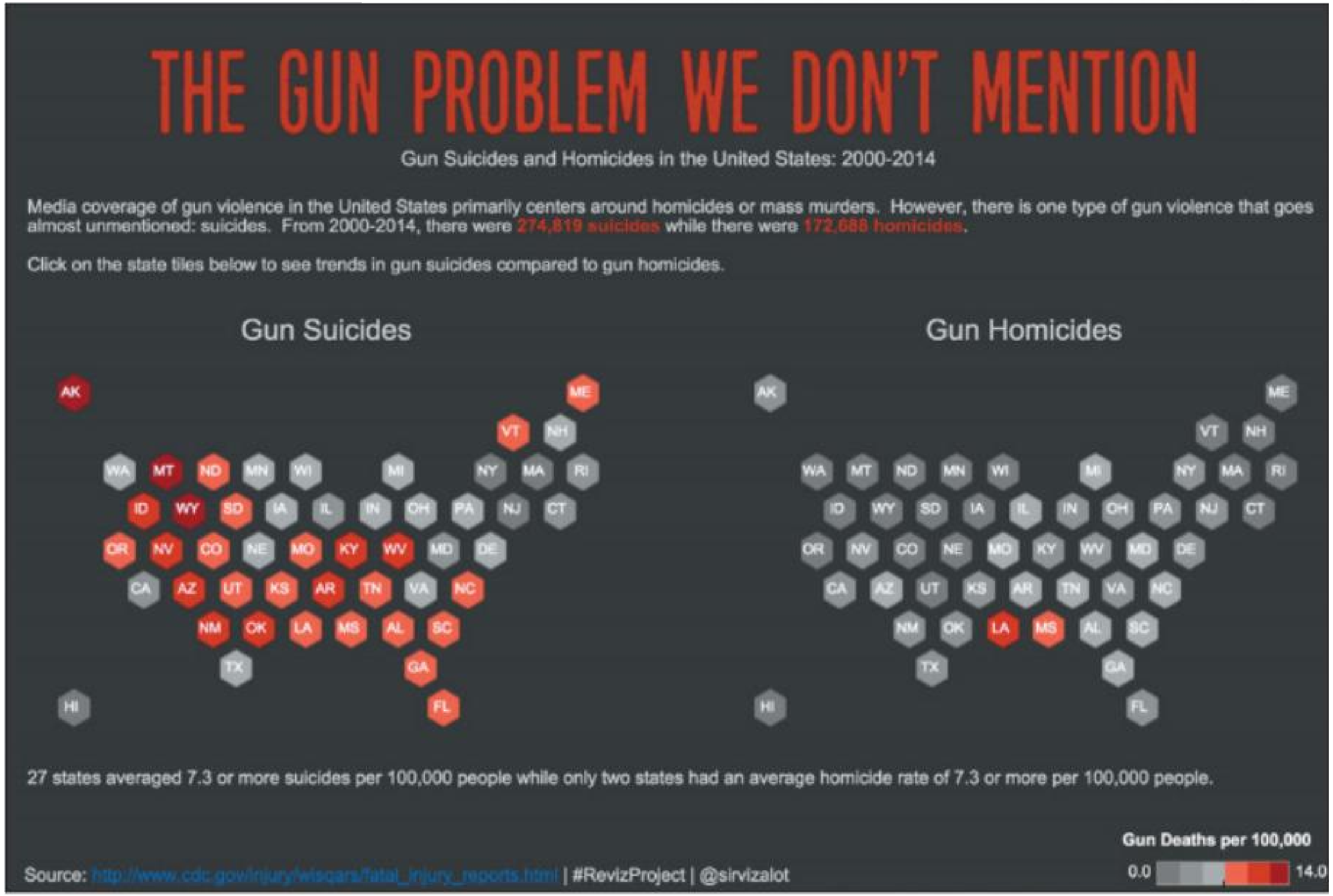
Data Visualization

While tag clouds summarize popular tags used in collaborative tagging applications, word clouds display statistics about word usage in a text collection. Here, a word cloud generated by the online generator at https://www.jasondavies.com/wordcloud/ shows the most frequent words in this chapter.



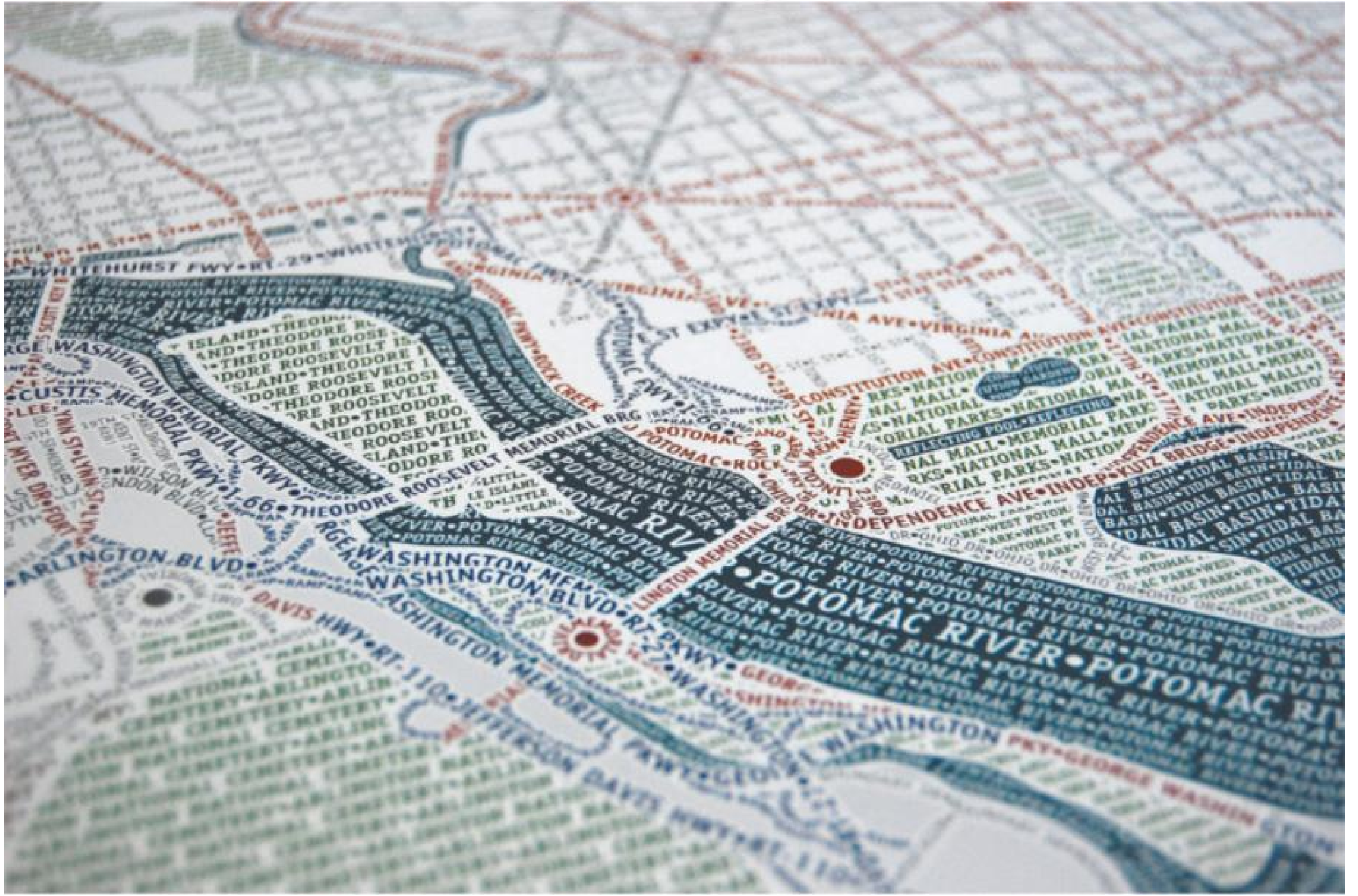
Data Visualization

Geographic visualization of gun deaths (suicides versus homicides) in the United States from 2000 to 2014 using data from the Center for Disease Control and Prevention (CDC). Instead of using the actual geographic boundaries of the individual states, this map replaces states with uniform hexagons that have been color-coded using the color scale on the bottom right. The benefit of this representation is to prevent large states from dominating the visual appearance of the overall map. The hexagons have been placed so that they largely preserve the topology of the original map. (https://public.tableau.com/profile/matt.chambers#!/vizhome/TheGunProblemWeDontMention/TheGunProblemWeDontMention)



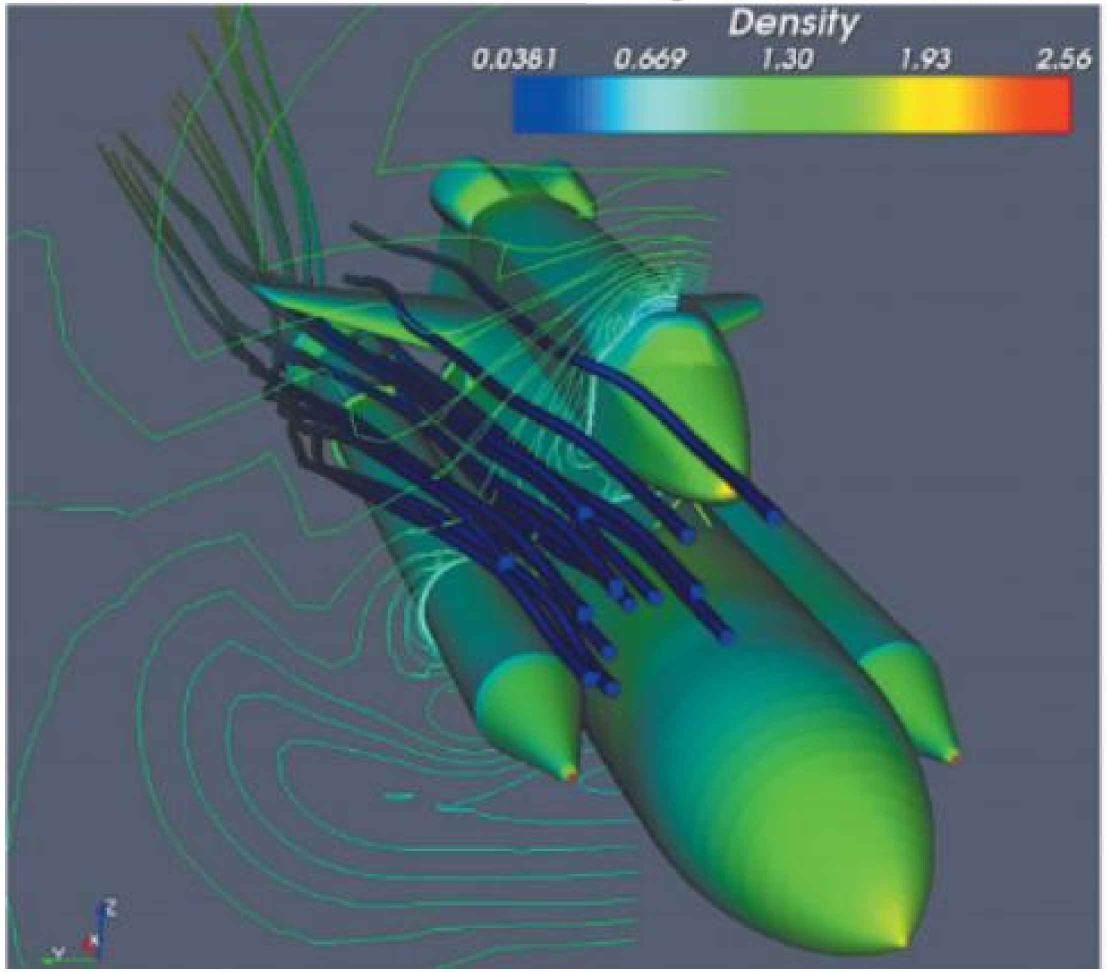
Data Visualization

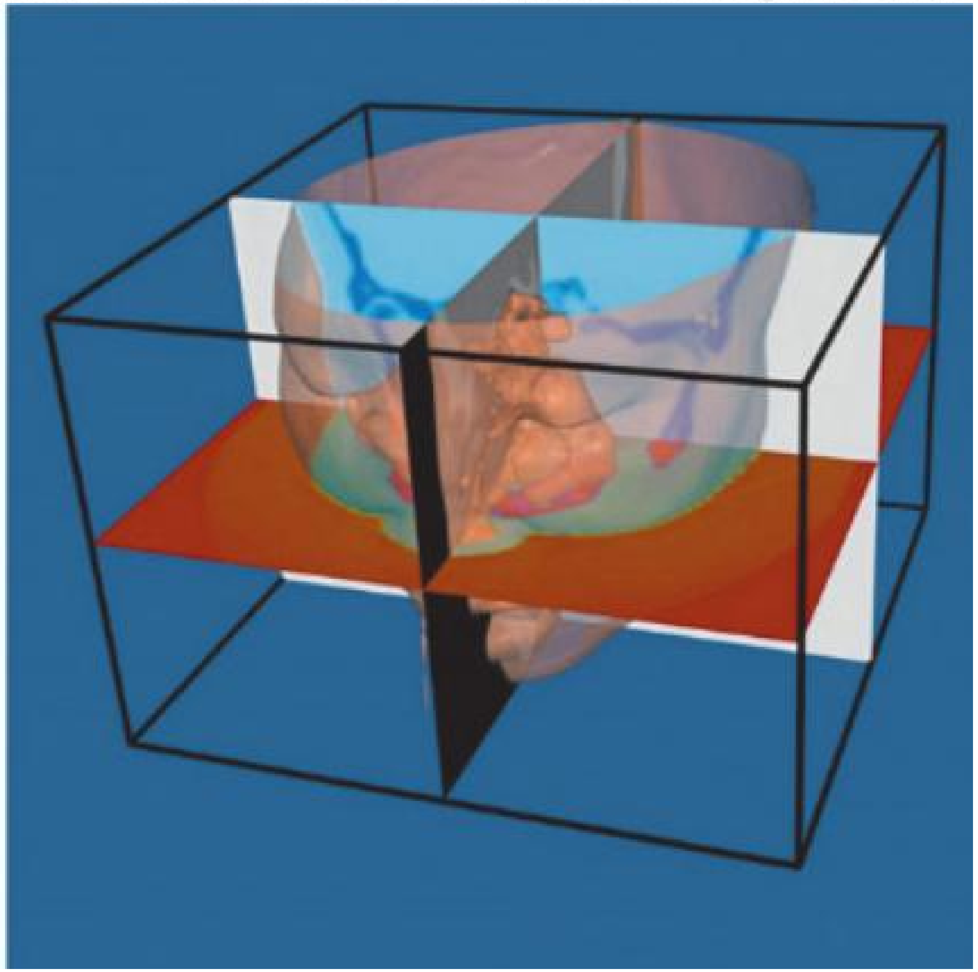
Typographic map of Washington, DC, created by Axis Maps. A typographic map consists entirely of text organized into shapes using colored labels of streets, parks, highways, shorelines, and neighborhoods. While this map took a skilled cartographer hundreds of painstaking hours to create, Afzal et al. (2012) later proposed an automatic approach taking mere minutes.



Data Visualization

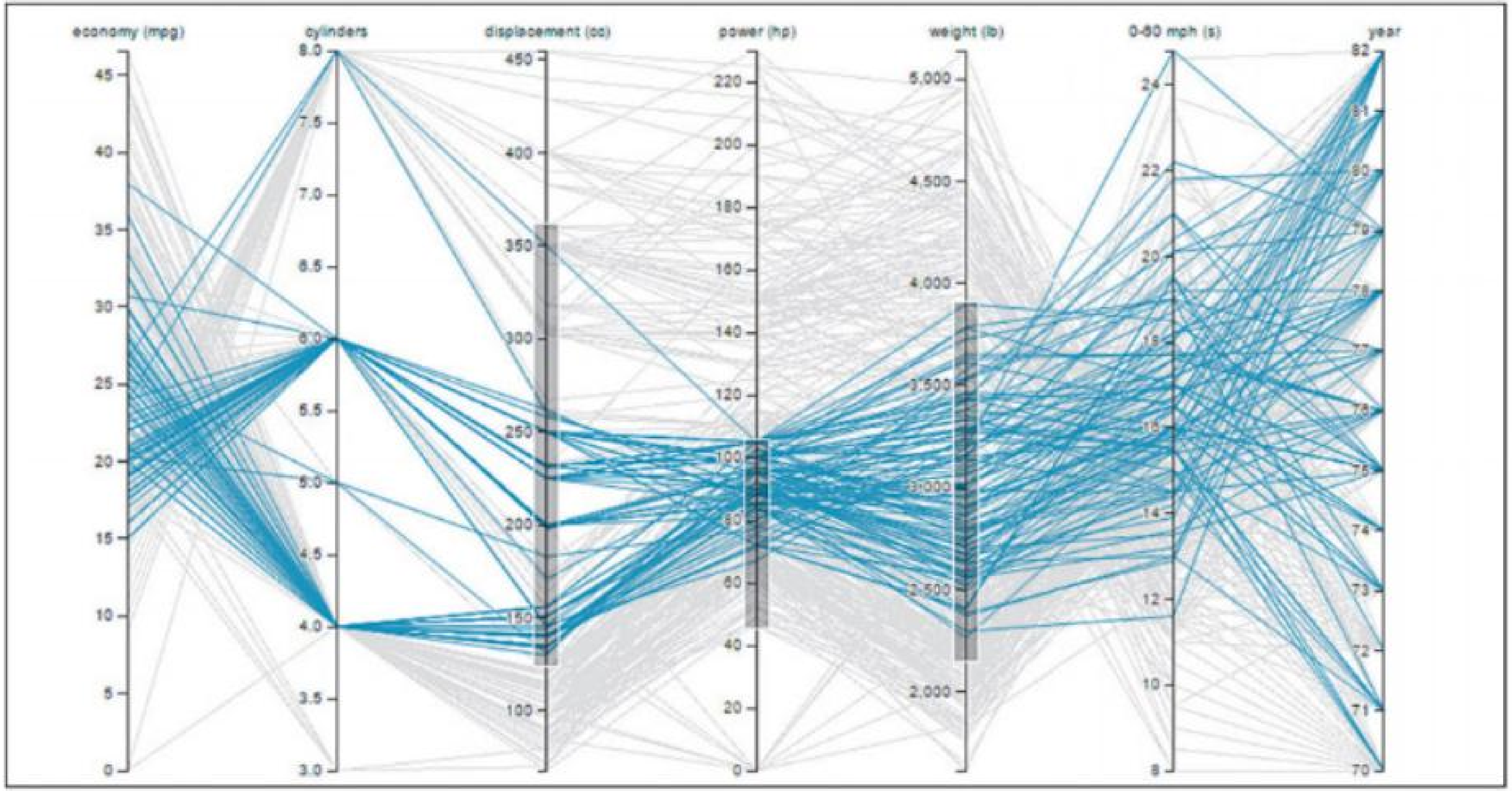
Two 3-D visualizations created using the Visualization Toolkit (VTK), a commercial software development library by Kitware, Inc. (http://www.kitware.com/). The left image shows flow density around the space shuttle using a rainbow color scale. The right image shows a CT scan of a human head with cross-sectional planes through the data.





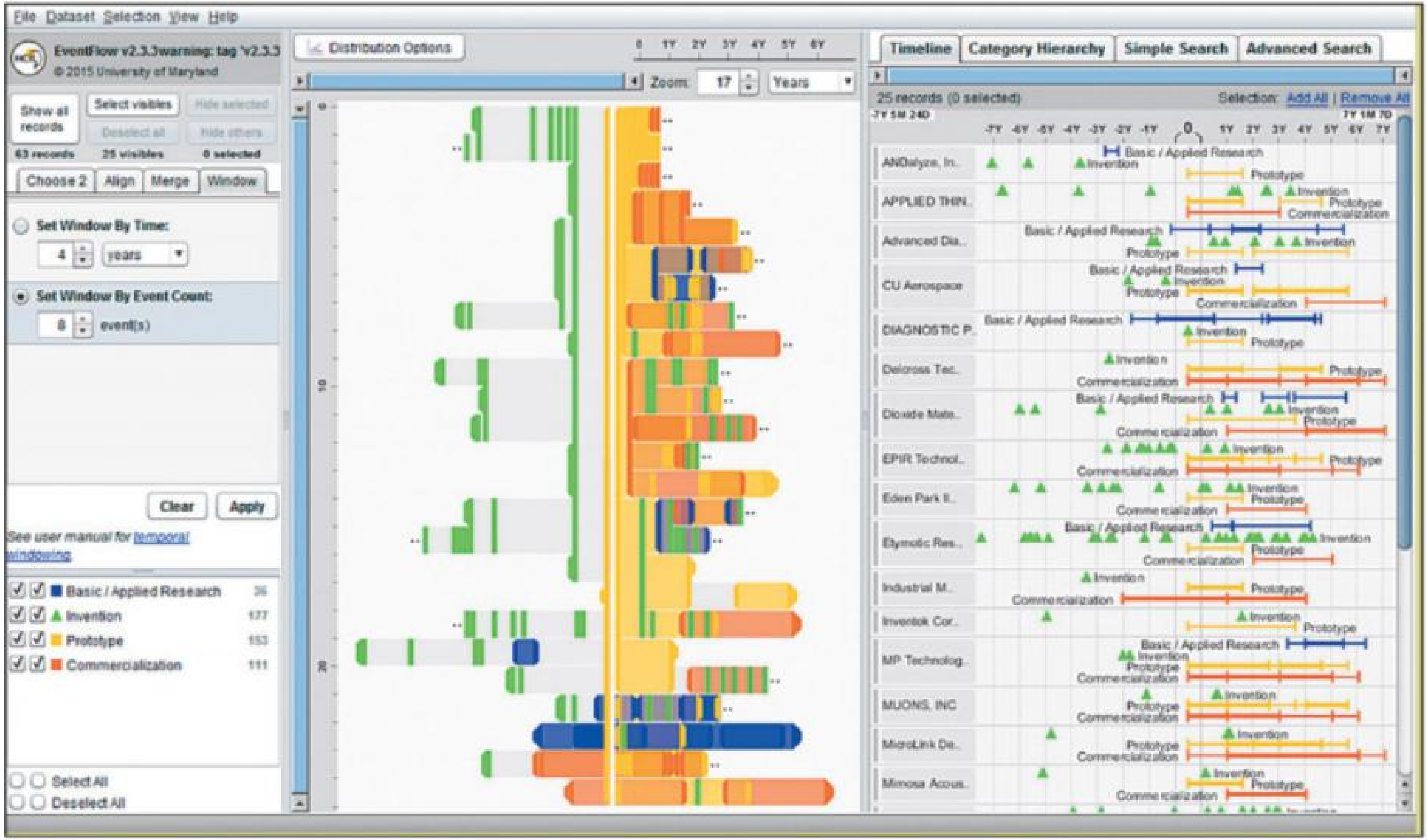
Data Visualization

Parallel coordinate visualization of cars from the 1970s and 1980s created using the D3 library. This visualization supports axis filtering where selecting data ranges on the dimension axes filters out the cars that do not meet all of the criteria (gray lines).



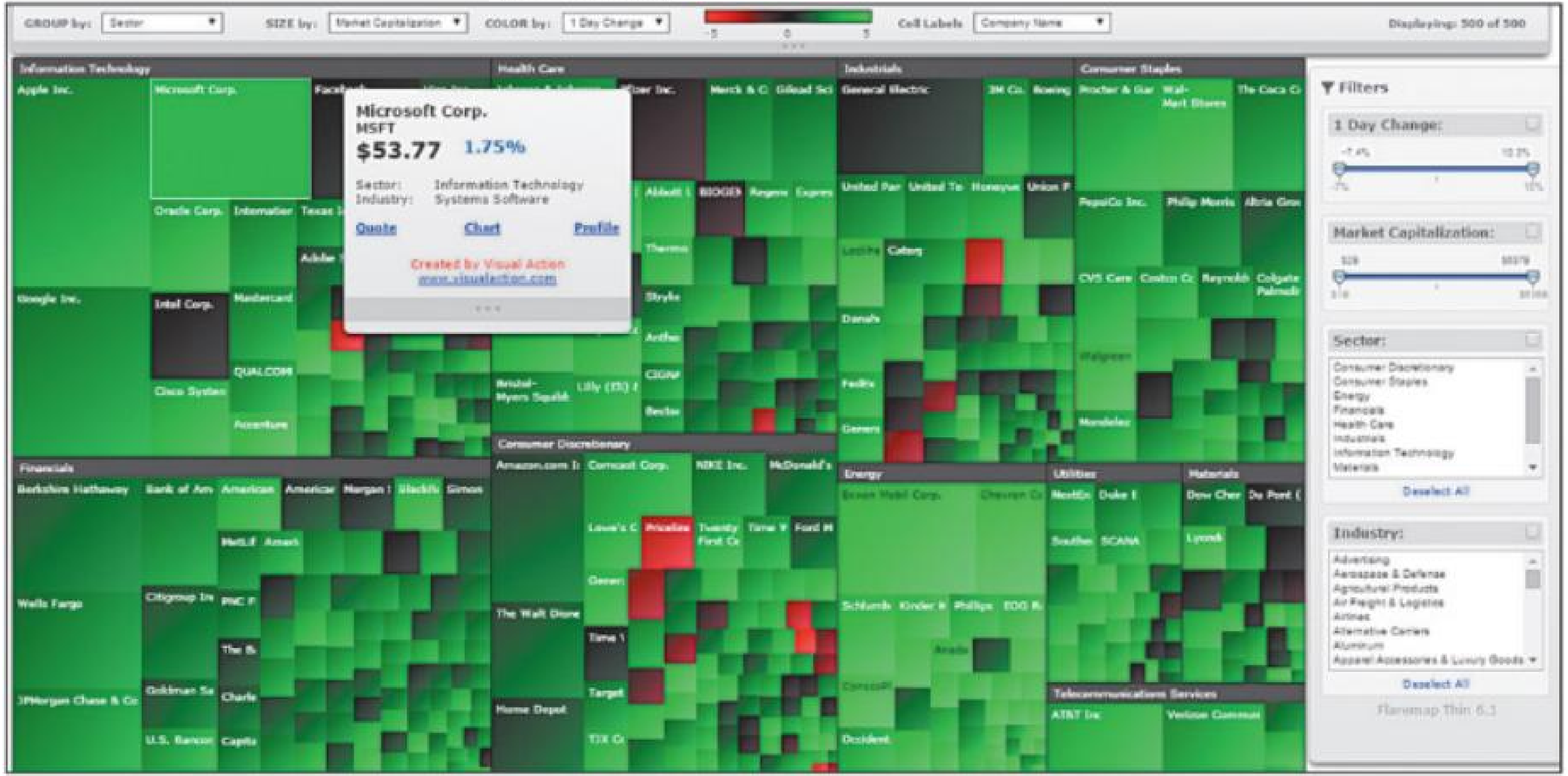
Data Visualization

The EventFlow (http://www.cs.umd.edu/hcil/eventflow) temporal event visualization system used to visualize sequences of innovation activities by Illinois companies. Activity types include research, invention, prototyping and commercialization. The timeline (right panel) shows the sequence of activities for each company. The overview panel (center) summarizes all the records aligned by the first prototyping activity of the company. In most of the sequences shown here, the company's first prototype is preceded by two or more patents with a lag of about one year between the last patent application and the first prototype.



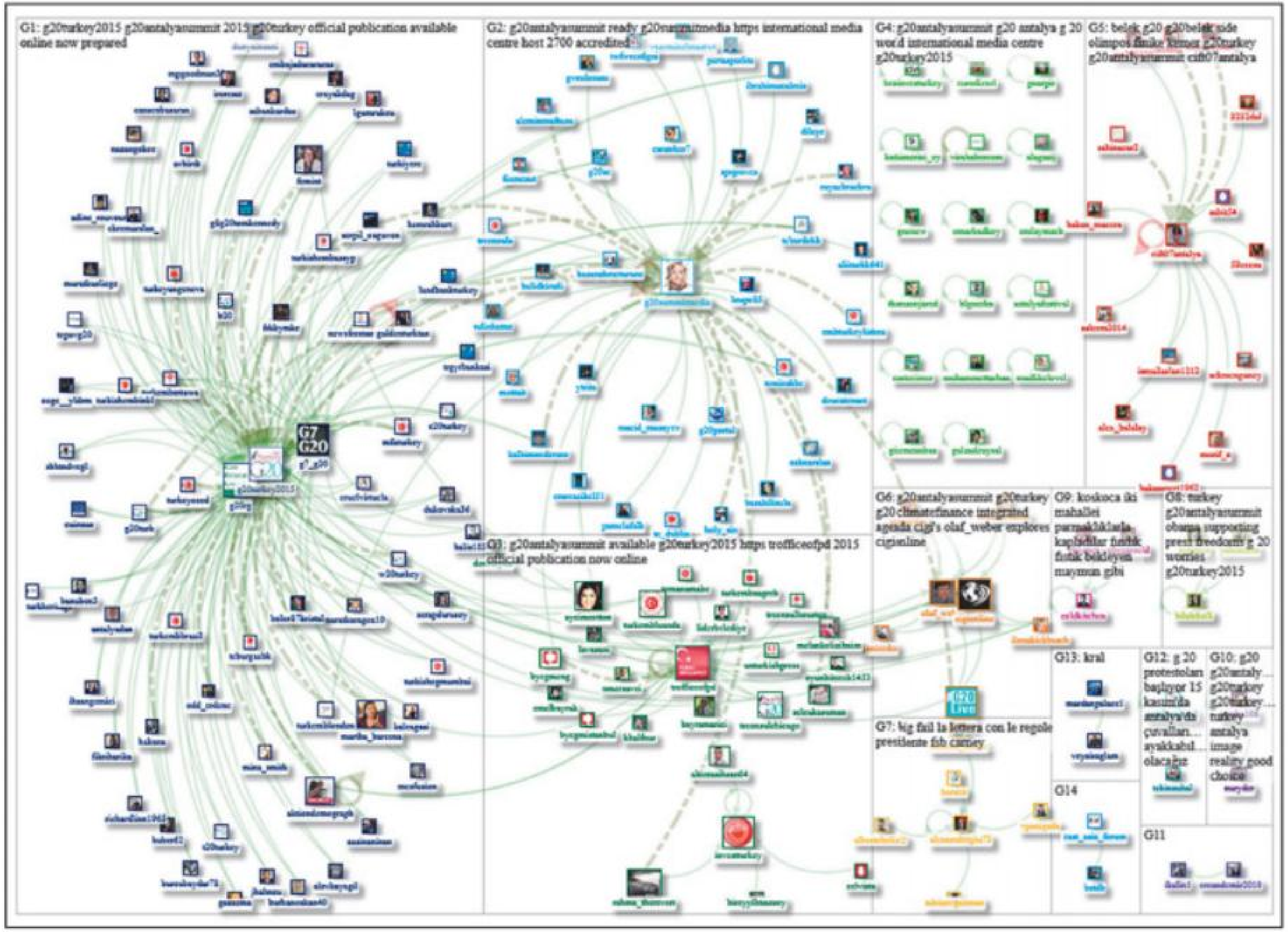
Data Visualization

The S&P 500 Market Monitor by Visual Action (http://www.visualaction.com/) a web-based treemap visualization showing stock performance of the 500 large companies making up the S&P 500 index on the NYSE and NASDAQ stock markets. Each rectangle represents a company, sized according to its market capitalization, colored based on its one-day change, and organized into sectors.



Data Visualization

Social network visualization built using NodeXL (Hansen et al., 2010) of 191 Twitter users tweeting with the hashtag "#G20AntalyaSummit" on November 9, 2015. The hashtag refers to the 2015 G-20 summit held in Antalya, Turkey, on November 15–16, 2015. The users have been grouped and laid out in boxes based on the contents of the tweets. NodeXL (https://nodexl.codeplex.com/) allows social scientists to collect, analyze, and visualize network graphs using a familiar interface that plugs into Microsoft Excel.



DISCUSSION

Q & A